Examining Leading Causes of Squirrel Interactions Based on Observed Environmental and Behavioural Factors

Group W13G06

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EXECUTIVE SUMMARY

This study aims to discover how environmental and behavioural factors can affect how squirrels interact with humans. This study involves The production of supervised learning models to predict associations between the variables aswell as clustering to analyse how the data naturally groups. Preprocessing of data and correlation analysis have also been used within the study to ensure results are as reliable and interpretiable as possible.

The accuracy results of the supervised learning models indicted that the models were unable to accurately predict how the environmental and behavioural features would impact interactions (KNN: 0.668, Random Forest: 0.395, Decision Tree: 0.365).

Correlation analysis indicated that behavioural features were the most influential in predicting interactions however were still not strong enough to allow for any significant conclusions to be drawn.

It was ultimately found that behavioural features were partially able to be used to predict squirrel interactions but were not significant enough to draw a valid and reliable relationship between the two.

1 INTRODUCTION

The squirrel and hectare datasets provide observations surrounding squirrel behaviours within central park in New York, collected from the 2018 Central park 2018 squirrel census. Within this study we aim to use the data contained within both sets surrounding the environmental features of the park and behavioural features of the squirrels to predict whether there is a relationship between said features and how the squirrels interact with humans.

The report will investigate the following research question:

Which environmental and behavioural factors most strongly influence squirrel behaviour towards humans, and how consistent are these findings across different modelling approaches.

To develop an analysis based on this question, we will first preprocess the data to merge the hectare and squirrel datasets, remove any outliers or unwanted features, handle missing data, and transform the data for the supervised learning models. We will then use correlation analyses to examine whether there are relationships between the environmental and behavioural features and squirrel interactions with humans and if so, which features correlate the strongest. Three learning models will be created (K-Nearest Neighbours, Decision Tree and Random Forest) to determine how well these features can predict squirrel interactions with humans and which is the best model for predicting this relationship. Finally, a clustering model will be used to discover if the features naturally cluster and provide a knew outlook into the data.

2 METHOD

2.1 Preprocessing:

The preprocessing component of the code began with merging two datasets. From the hectare.csv file, 'Hectare', 'Shift', 'Litter', 'Hectare Conditions' and 'other animal sightings' were merged into the squirrel.csv files dataset. A left join was used to merge the 'Hectare' and 'Shift' columns to ensure that any missing hectare data would be entered as null rather than removed. This allowed us to inspect the null results later in preprocessing rather then have them removed during the merge. This merge allowed for a singular set of data that had a multitude of environmental factors which could later be used to predict squirrel behaviour.

Once merged, a print() statement was used to create a report of 'Nulls introduced by merge'. Within this report, it was found that the litter column of the dataset returned 1595 null values, which was equal to 52.8% of the records. Thus it was decided that the litter feature was unreliable and would not produce any meaningful results in relation to squirrel behaviour and was removed from the dataset using a df.drop() statement. Furthermore, 156 records with missing 'Hectare conditions' were removed using a df.dropna() statement. This lowered the overall records from 3023 records to 2860, which was a 5.3% reduction and was considered appropriate as 'Hectare conditions' was an important feature within the analysis. The 111 null records found within the 'Animal sighting' feature were assigned to a value of 0 using na=False. Any sightings including a dog was assigned to the feature 'Dog_nearby' and the rest were also assigned to 0. This was due to dogs being the most common animal sighting and thus the most reliable animal influencing squirrel interactions.

The variable that was the focus of the model, 'Interaction', came from three different columns in the squirrel.csv dataset 'Approaches', 'Run from' and 'Indifferent'. The observations with 'Indifferent' as TRUE were assigned to 0, 'Approaches' was assigned to 1, and 'runs from' was assigned to 2, with a priority ordering being used within the code to ensure that if, for example, both 'indifferent' and 'approaches' were true, the overall observation would be assigned to 1. This created a single 'Interaction' column which could be later used as a basis for the predictive supervised learning models.

The 'Hectare conditions' observations contained 5 different categories, 'Busy', 'Calm, Busy', 'Moderate', 'Medium' and 'calm'. These were encoded into a new 'Busy_stress' data column which mapped the observations onto an ordinal scale, with 2 being 'Busy' and 'calm, Busy', 1 as 'moderate' and 'medium' and 0 being 'calm'. Additionally a 'for' function was used to map binary values to the True/ False values

of the observed squirrel actions ‘Running’, ‘Chasing’, Climbing’, ‘Eating’, ‘Foraging’, so the model could accurately interpret the data. Any missing values were assigned to 0 (False).

2.2 Correlation Analysis:

To examine the correlation between each interaction and feature, we used a point-biserial correlation between each feature and a binary indicator for each interaction type. This was used as our data was not continuous or normally distributed. Therefore, rather then a heatmap, we created individual bar plots for each interaction type showing how strong the correlation between each feature was with the interaction. On top of our data being discrete, we also used this method as it was believed that a single correlation graph would produce uncertain results, as each interaction type was likely influenced differently by the different features. It is important however, to note that correlation measures associations and not causation, and interactions between the two features are not directly captured.

Bar charts were then created which showed the mean feature values for each interaction. This allowed for easier visualization of the correlations between the features and interactions, improving the interpretability of the patterns.

2.3 Supervised Learning:

To find the best predictive supervised learning model for our data set we used three different models, which were ‘Random Forest’. ‘Decision Tree’ and ‘K-Nearest Neighbours (KNN)’. Random Forest was used as it is a flexible learning model that produces a number of decision trees and combines the predictions from each of those individual trees to produce a more accurate prediction. It also reduces the risk of overfitting the data as the process of taking the average of all the decision trees lowers both prediction error and variance. The Decision Tree was used as it is a more interpretable learning model than the Random forest model, allowing for easy observation of how the features may impact interaction types. Finally KNN was included as a distance based learning model which makes predictions based on similarity, looking at a specified number of observations within the training model and making a prediction based on whichever interaction type is the most common.

The dataset was split into 80% training and 20% test sets to ensure that the models have enough data to make a reliable prediction. Furthermore, Stratified sampling was used to ensure that the distribution of features was equal across subsets. This was important as the interactions data was unbalanced, with ‘Approaches’ making up only 5.9% of the overall interactions and thus needed to be reflected by the learning models.

The large imbalance in the Interaction data was accounted for by using ‘class_weight=’balanced’’ in both the Random Forest and Decision Tree models. This adjusts the effect of the minority class, ‘Approaches’, so that the models did not predict the majority class ‘Indifferent’ each time in the test set. Unlike the other two models, however, we could not balance the classes within the KNN model which accounts for a much larger difference in accuracy and prediction results compared to the other two models. However, z score standardization was applied to the data before the KNN training as it the model is sensitive to feature scale.

It is important to note that a value of k=3 was selected for KNN which represents a possible limitation for the analysis. Furthermore, for random forest, ‘n_estimators=100’which could also be a possible limitation for the analysis.

To compare the supervised learning models performances, normalized confusion matrices were created to show the proportion of accurately classified interaction types by each model. A bar chart comparing the accuracy of each models performance was also created which gave an image of the overall correctness of the models performances

2.4 Clustering

Clustering was used to determine whether there were natural groupings within the dataset. Within the analysis, K-Means clustering was used. Before clustering all features went through z-score standardization to ensure that the clusters reflected the true nature of the observed behaviours.

An Elbow Method lineplot was produced, mapping the distortions across values of k. The values of k mapped was from 2 to 8 and the optimal k value found was k=5 as this was when the curve began to flatten. After selecting the optimal k, the mean value of each feature was calculated for each of the clusters. This was used to discover which behaviours were most influential in each cluster and whether they could have had an effect on interaction behaviours.

3 RESULTS AND ANALYSIS

3.1 Preprocessing Results and Analysis

Data preparation had a substantial impact on both model performance and clustering results. Merging the squirrel dataset with the hectare-level environmental dataset resulted in considerable data gaps, especially for the Litter feature with 1595 missing entries. Given that over fifty percent of its entries were missing, and the other two had far fewer missing values, this variable was excluded to avoid introducing bias and unreliable imputation. Accordingly, the resulting dataset focused on environmental stress indicators, canine presence, and observed squirrel behaviours.

Table 2: Missing values introduced after dataset merge

```
Nulls introduced by merge:
Litter                   1595
Hectare Conditions        163
Other Animal Sightings    111
dtype: int64
Data integration successful. Total records: 2860
Interaction
Indifferent    0.724825
Runs from      0.215734
Approaches     0.059441
Name: proportion, dtype: float64
```

Further, categorical variables were converted into numerical equivalents to enable processing by the chosen learning models. Environmental variables were encoded into a Busy_Stress scale, which reflected progressively higher levels of environmental stress, whereas behavioural variables such as Running, Chasing, Climbing, Eating and Foraging were converted to binary features (1 = present and 0 = absent) using binary label encoding. Owing to its anticipated influence on avoidance behaviour, presence of dogs was also represented as a binary feature.

Standardisation was applied to variables used as inputs to distance-based methods, specifically to KNN and K-Means clustering, since features with larger numerical ranges are most likely to dominate distance calculations and introduce bias into the learning process. Once standardised, all the variables contributed equally to distance-based comparisons, ensuring greater consistency in model performance.

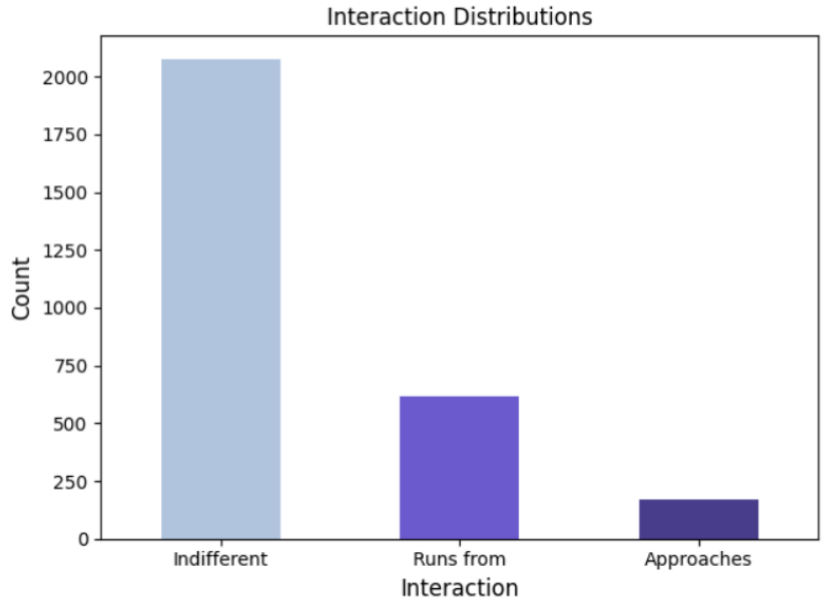


Figure 1: Distribution of Squirrel Interaction Outcomes

As shown in the figure, preprocessing further exposed an imbalanced distribution of interaction outcomes, and the Indifferent outcome was observed most frequently (~72%), with Runs from (~22%) and Approaches (~6%) occurring at lower frequencies. Class imbalance is known to affect classification performance and tends to result in a lower predictive reliability for minority classes. This imbalance is therefore a critical factor in the interpretation of the performance of supervised models.

These preprocessing steps established a dataset suitable for both classification and clustering tasks and ensured that the results were not biased by missing data, inconsistent feature scales, or unprocessed categorical variables

3.2 Correlation Analysis Results

Correlation analysis was performed to explore the associations between the environmental variables, squirrel behaviours and the interaction outcomes

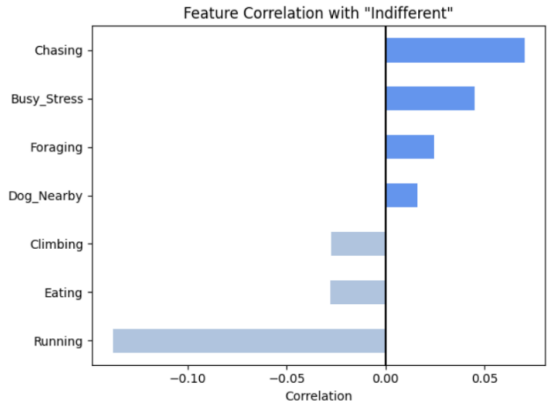


Figure 2: Indifferent correlations

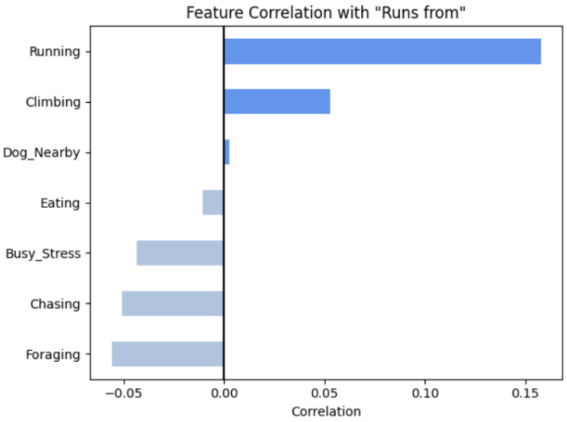


Figure 3: Runs From Correlations

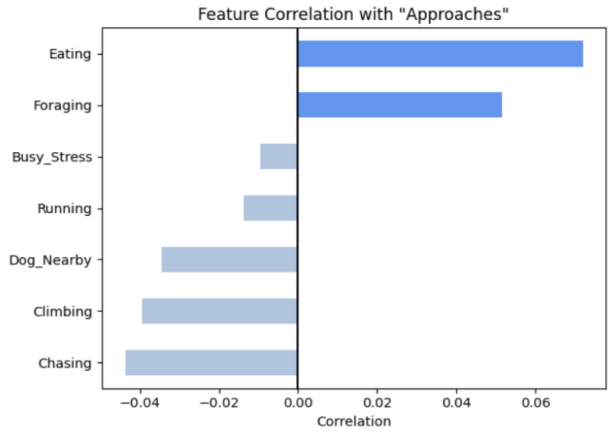


Figure 4: Approaches correlation

Figures 2-4 clearly show the correlation between each predictor, and the three interaction classes, where several meaningful patterns are apparent.

With respect to the Approaches class, positive correlation is evident for Eating, Foraging and Climbing variables. These behaviours indicate active or feeding squirrels, implying that individuals engaged in resource acquisition exhibit a greater propensity to approach humans. Conversely, the canine presence shows a negative correlation, suggesting that the presence of potential predators inhibits approach behaviour.

In the Runs from class, the strongest positive correlation is observed for Running which illustrates that squirrels engaged in running behaviour are more likely to be moving away from humans. Climbing also shows a weak positive correlation, supporting this interpretation as both are escape related behaviours. In contrast, Foraging, Chasing and Busy_stress show weak negative correlations, suggesting that flight responses are less common in stationary feeding contexts or in highly congested areas. Notably, Dog_Nearby shows near zero correlation in the linear analysis.

However, the Indifferent class shows weaker correlations across all features ,which implies that neutral behaviour occurs across a wider set of environmental conditions. These trends are further shown in Figure 5, presenting a comparison of average predictor values among the interaction categories.

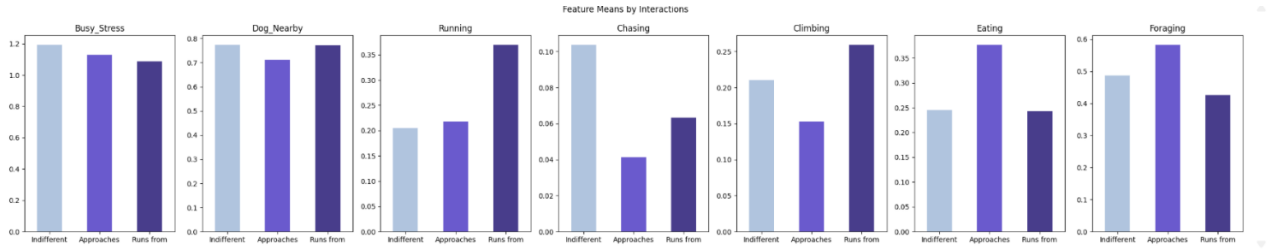


Figure 5: Mean predictor comparison chart

Squirrels that approach humans display higher rates of feeding and investigatory movement, whereas those that run away are associated with heightened environmental stress and canine presence.

Overall, the correlation analysis demonstrates that the environmental context and the behavioural activity are strongly linked to the interaction between squirrels and humans which directly addresses the research question.

3.3 Supervised Learning Results and Analysis

Performance of models was evaluated using accuracy, precision, recall, classification reports and confusion matrices. Collectively, these measures offered complementary perspectives on model behaviour since accuracy alone is known to be misleading when the dataset is imbalanced.

Three models were compared: Random Forest, Decision Tree, and K-Nearest Neighbour (KNN).

Random Forest

	precision	recall	f1-score	support
Indifferent	0.75	0.41	0.53	415
Approaches	0.09	0.56	0.15	34
Runs from	0.28	0.29	0.29	123
accuracy			0.40	572
macro avg	0.37	0.42	0.32	572
weighted avg	0.61	0.40	0.46	572

Decision Tree

	precision	recall	f1-score	support
Indifferent	0.76	0.36	0.49	415
Approaches	0.09	0.59	0.15	34
Runs from	0.27	0.31	0.29	123
accuracy			0.37	572
...				
accuracy			0.67	572
macro avg	0.33	0.35	0.34	572
weighted avg	0.59	0.67	0.63	572

Table 2: Model accuracy Comparison

Table 2 summarises overall model accuracy. As shown in figure 7, KNN achieved the highest raw accuracy but figure 8 reveals that this was driven by a majority-class prediction that KNN failed to predict the Approaches class entirely. Even though Random Forest and Decision Tree achieved lower raw accuracy, they produced non-zero recall across all three classes, making them more suitable for the research question.

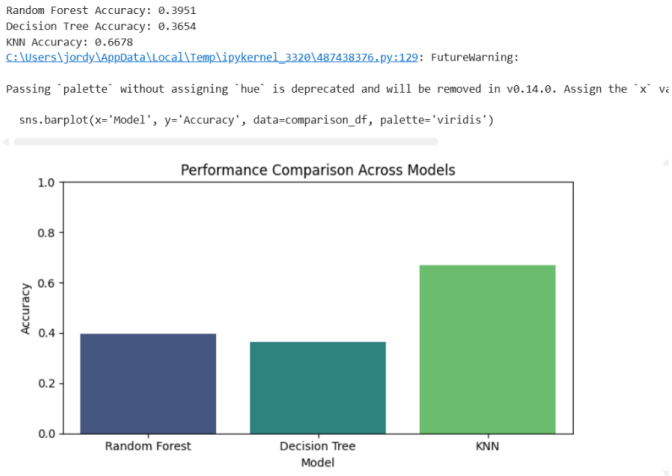


Figure 7: Model accuracy comparison

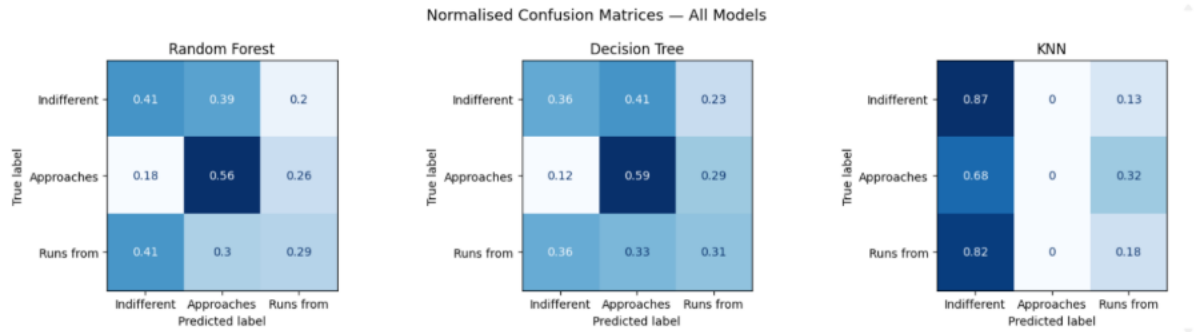


Figure 8: Normalised Confusion Matrices

The confusion matrices in Figure 8 offer deeper insight into the model's performance. The Indifferent class is classified most consistently with the highest accuracy across all models. Classification errors occur more predominantly between the Approaches and Runs from classes indicating distinguishing between these behaviours is more challenging. This pattern suggests a degree of overlap in environmental conditions where squirrel responses are not uniformly predicted.

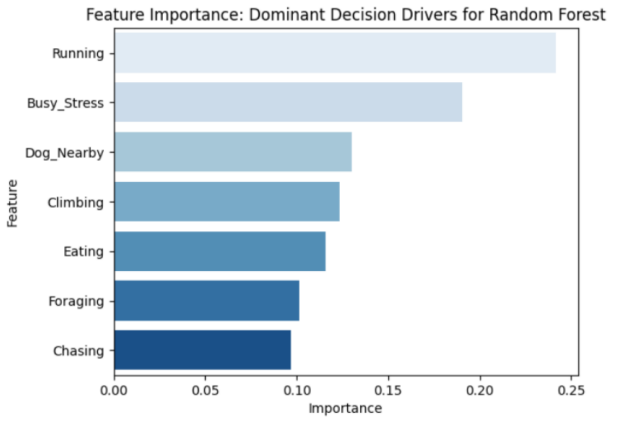


Figure 9: Random Forest Feature Importance

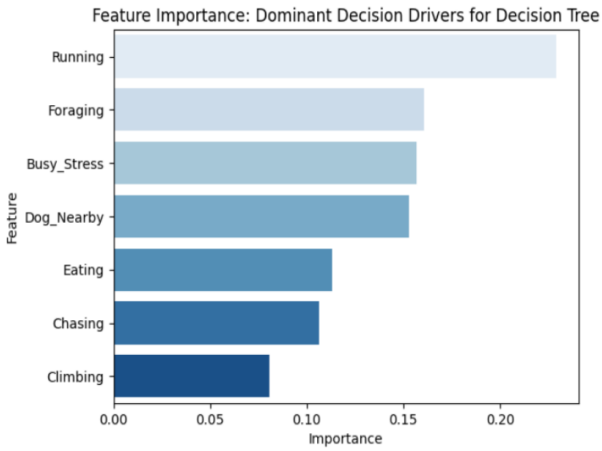


Figure 10: Decision Tree Feature Importance

Further, Feature importance analysis in Figure 9 and Figure 10 offers additional insight into model decisions. Both the Random Forest and Decision Tree models identify Running as the most dominant predictive variable but activity-related features such as Eating, Foraging, and Climbing also contribute substantively to predictive performance

These results directly support the research question by identifying which environmental conditions are most strongly associated with squirrel interactions.

3.4 Clustering Results and Analysis

Clustering analysis was performed using K-Means to investigate whether natural behavioural groupings exist independently of the interaction class labels.

The elbow method as shown in Figure 11 was used to identify the optimal number of clusters which indicated that data could be meaningfully partitioned into multiple behavioural categories. Based on our evaluation, k = 5 was chosen as the optimal number of clusters.

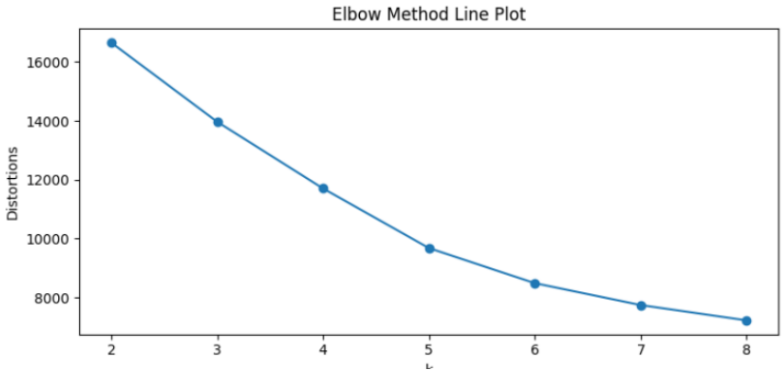


Figure 11: Elbow Method Line Plot

Average Feature Values (Clusters):

Cluster	Busy_Stress	Dog_Nearby	Running	Chasing	Climbing	Eating	Foraging
0	1.249	0.802	0.081	0.0	0.061	1.000	0.539
1	1.103	0.800	0.188	0.0	1.000	0.110	0.147
2	1.140	0.771	1.000	0.0	0.000	0.078	0.233
3	1.156	0.735	0.000	0.0	0.000	0.000	0.811
4	1.188	0.762	0.291	1.0	0.165	0.115	0.207



Figure 12: Cluster feature averages

Figure 12 shows the average feature values across the clusters, which revealed clearly distinct behavioural profiles. Certain clusters exhibit elevated levels of environmental stress and dog proximity corresponding to environments where avoidance behaviour is likely.

Conversely, other clusters demonstrate higher levels of feeding and activity behaviours, representing squirrels with greater tolerance of human proximity. These outcomes suggest that squirrel responses are distributed along a behavioural continuum rather than forming perfectly separated groups.

The findings hence align with the supervised learning results where classification occurs primarily between Approaches and Runs From classes. Taken together, the consistency observed between supervised and unsupervised learning strengthens the conclusion that observablee behaviours, environmental stress and predator presence jointly shape of squirrel interaction outcomes.

4 Discussion

The research question of this report was whether squirrel behaviour towards humans can be predicted by an array of environmental and behavioural factors. Based on the results formed from producing a correlation analysis, the use of a range of supervised learning models and clustering, it was found that squirrel behaviour towards humans can only partially be predicted by environmental and behavioural observations and thus indicates that much further research is needed when attempting to uncover what drives squirrel interactions with humans.

The correlation analysis highlighted that no feature was strongly associated with squirrel interactions with humans, with the ‘Running’ behavioural feature having the highest correlation with any interaction with humans, which was ‘Runs away’ and was approximately 0.16. This indicates that predicting how a squirrel will interact with a human based on the dataset of observations would be difficult as the weak individual correlations indicate no feature specifically drives one of the three interactions. However, the ‘Feature Means by Interactions’ correlation graphs suggest some interpretable patterns. The most influential results surround the behavioural features rather than the environmental features. The patterns indicate that if a squirrel is already running or climbing, it will run from humans, if a squirrel is chasing something, it will remain indifferent to humans, and if a squirrel is eating or foraging, it is more likely to approach a human. This suggests that a squirrels current behaviour correlates with how they interact with humans far greater than the conditions around them.

The results generated by the supervised learning models further indicate that the behavioural and observational features are not the strongest influence on squirrel interactions with humans. The ‘Performance Comparison Across Models’ graph shows a 39.5% accuracy for Random Forest and a 36.5% accuracy for Decision Tree which would likely have been exceptionally higher if the features were actually predictive of interaction. The KNN model returned a 66.8% accuracy which was deceptively high when compared to the other models. This was explained in the confusion matrices as the KNN model completely ignored the ‘Approaches’ interaction, which was only 5.9% of the datasets observed interactions and was due to KNN being unable to handle a large class imbalance in data.

The clustering analysis provided a different approach to looking at how interactions between squirrels and humans were influenced by environmental and behavioural factors. After producing an elbow method graph

it was found that the optimal k value was k=5 rather than k=3, implying that the data naturally clusters into five groups rather than the three already defined interaction groups. The average feature means uncover that the clusters revolve more around the squirrels behaviour rather than their interactions with humans. This helps to reinforce the results from the supervised learning models as the clustering around behaviour suggests that the relationship between environmental and behavioural factors and interaction with humans is not strong enough for the features to cluster around the three separate interaction groups.

5 Limitations and Reflection

The dataset and method to analyze the research question both posed some significant limitations that could have influenced the results. The most significant limitation within the dataset was the unbalanced interactions data, with only 5.9% of observations accounting for 'Approaches'. This significantly affects the supervised learning model predictions as it diminished the possibilities for reliable predications within any of the prediction models. A more balanced dataset with a greater amount of 'Approaches' observations could drastically change the results and create a meaningful improvement on the performance of the learning model predictions. Furthermore, the 'Litter' environmental factor was excluded from the analysis due to 52.8% of the values being missing and presenting as 'Null'. This directly influenced the results and explains why the behavioural observational factors produced more significant results than the environmental observational factors. A Hectare dataset that was more complete would provide greater context to environmental factors driving squirrel interactions and this improved the learning models predictions.

Within the method used to create the supervised learning models, there were limitations found in the number of neighbours for the KNN and number of trees for Random Forest. For KNN, k=3 was used, however, this was based on minimal foundation and testing to find what the best k value should have been implemented, as this could have improved the predictions slightly. The same can be said for the number of trees in Random Forest (100). If testing to find the best nmber of trees was implemented a more accurate result may have been predicted. Furthermore, the class imbalance was not accounted for in the KNN learning model and thus rendered the results from that predictive model insignificant. In future further research should be done into what values are most optimal for the dataset being used.

6 Conclusion

This study was ultimately driven by the idea of discovering whether there was enough association between observed squirrel interactions with humans and their environment and behaviours. The findings found that whilst squirrel behaviour was partially able to be predicted by specific supervosed learning models, there was not enough of an association between the two variables to make a reliable and accurate prediction. Across all the models produced within the study it was found that the behaviour of the squirrels at the time of observation had the highest probability of assisting the prediction of their interaction with humans and thus, illustrates that future research should be focused around bheaviours only rather than on environmental factors.

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Data/vfnx-vebw